

Bioinformatics for engineers

Genome Annotation

Msc. Juan Sebastian Malagón Torres

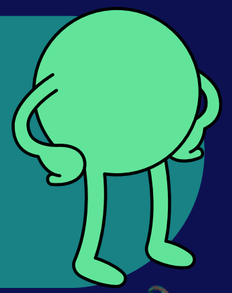
Teaching assistant. Eliana Zuluaga Aponte

Universidad Javeriana

Pontificia Universidad JAVERIANA Colombia

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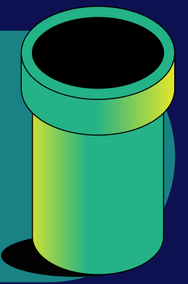
2. Structural annotation



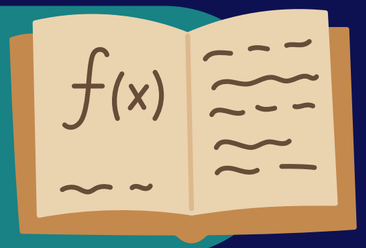
3. Prokaryotes vs. Eukaryotes



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WHAT DOES IT REALLY MEAN TO HAVE THE COMPLETE SEQUENCE OF AN ORGANISM?

From raw data to valuable information

So far, we have focused on understanding and manipulating the bioinformatics tools needed to obtain, process, assemble, and represent genetic information.

BUT...

How can we know what ***information*** the genome of an organism gives us? What relevant genes can we find, what functions do they perform, which fragments of the genome are not coding, are there repetitive elements?



Say hi to genome annotation

It is the process of ***identifying*** and ***describing*** the location, ***structure***, and ***function*** of elements present in a genomic DNA sequence using computational tools and biological databases.

Understanding metabolism



Locating Open Reading Frames



Defining gene architecture



Understanding the "Ontology" of genes



Identifying variants (SNPs, indels...)



For making this happen we´ll have to work with coordinates (where we find relevant information) and metadata (data about the data).

STRUCTURAL ANNOTATION

Working with genomic coordinates

What types of elements we need to ***map*** in a genome?



- *Genes*
- *Regulatory regions*
- *ncRNA*
- *Repeats*
- *Pseudogenes*
- *Paralogs*
- *Exons and introns**

Why do we need structural annotation?

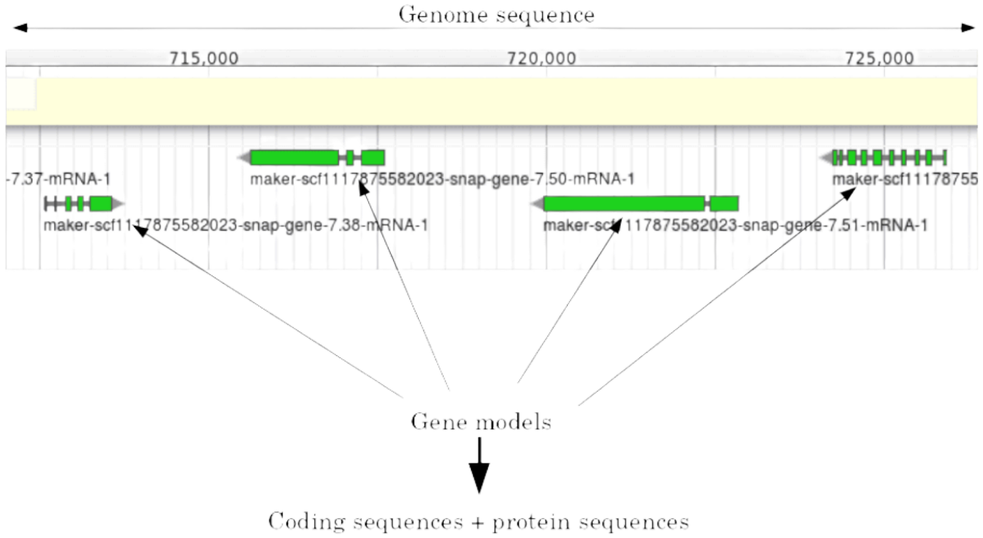
Genome sequence (raw entry)

↓

Gene models (structural interpretation)
- Exons and introns (in eukaryotes) and directionality

↓

Coding* and protein sequences

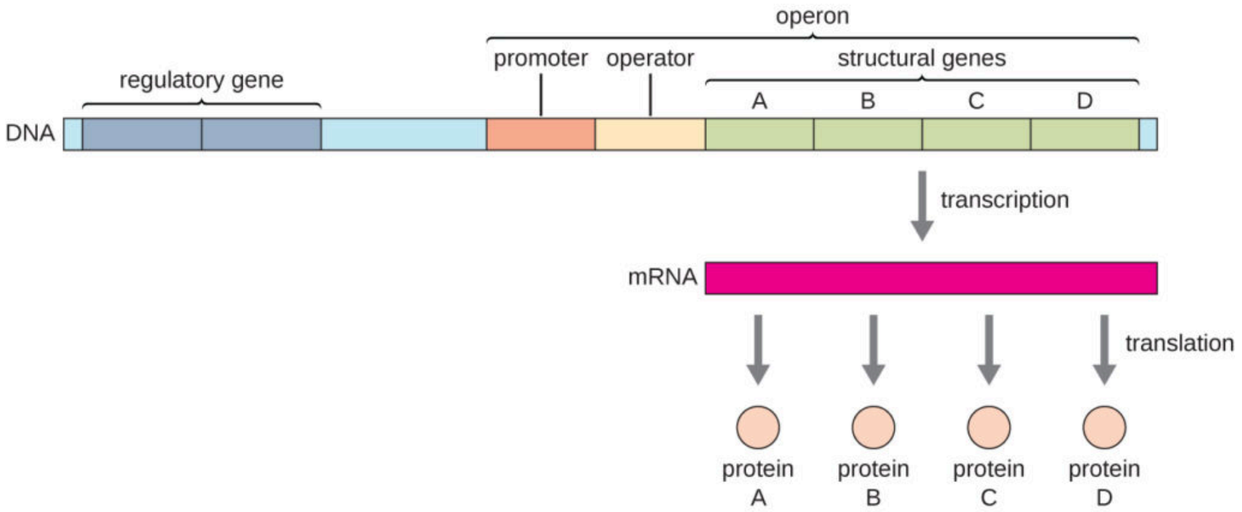


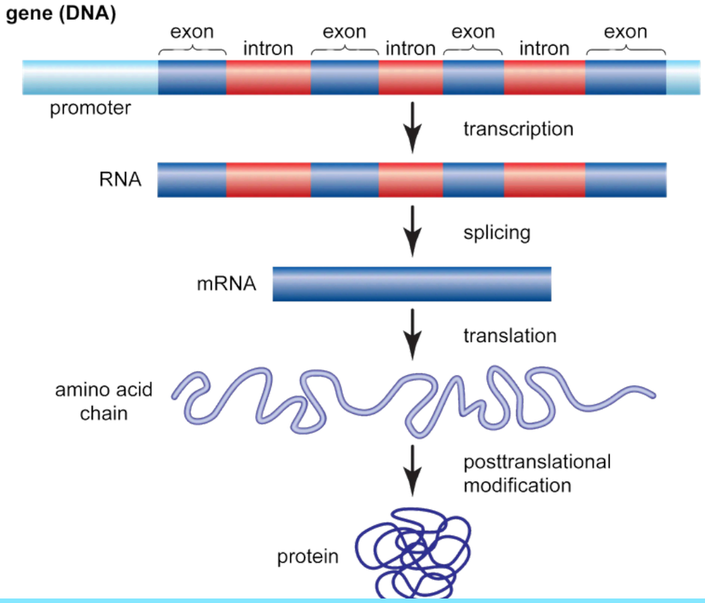
Compare with other species:
- Presence/abscense/mutations of genes
- Family reduction or expansion
- Structural variants

Expression and metabolism
- Map complete metabolic pathways
- RNA-seq

DOES THE STRATEGY CHANGE BETWEEN PROKARYOTES AND EUKARYOTES?

Prokaryotic vs. eukaryotic genes





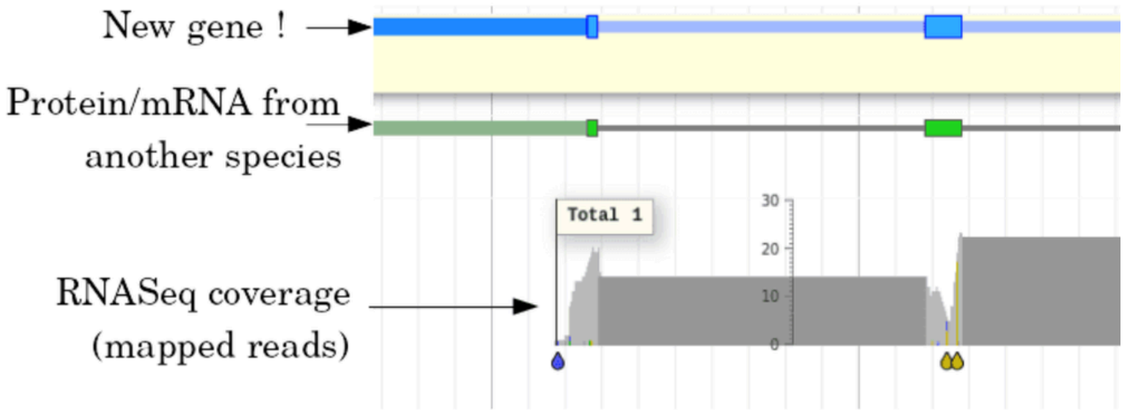
- *Their genetic structure is preserved*
- *Promoters with region -35 and TATA box.*
- *Polystronic operons.*
- *No splicing.*
- *Shine-Dalgarno marks the start of translation.*

- *Genes with exons/introns.*
- *Alternative splicing (GT/AG signals).*
- *Complexity: a single region can generate multiple isoforms.*
- *Low coding density*

How to make a structural annotation happen

Automatic annotation (not ideal)

- DNA signals are short and unpredictable motives
- *The software cannot trust only sequences, it needs evidence.*



Types of evidence

- **RNA-seq:** RNA reads that align with a region of the genome (the region is *expressing* itself)
- **ESTs or Transcripts:** *Already known RNA sequences* of the same species or very close species.
- **Protein Alignment:** If a known protein from another species aligns well with a region of your genome, there is very likely a *homologous gene*.

Limitations for new, very distant or unexpressed genes.

Sometimes you just don´t have the evidence, so you refine your prediction methods...

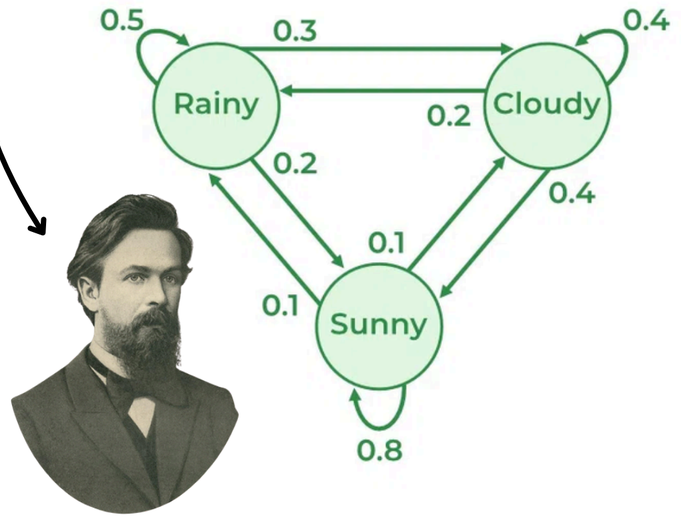
Ab initio prediction

Prediction based on:

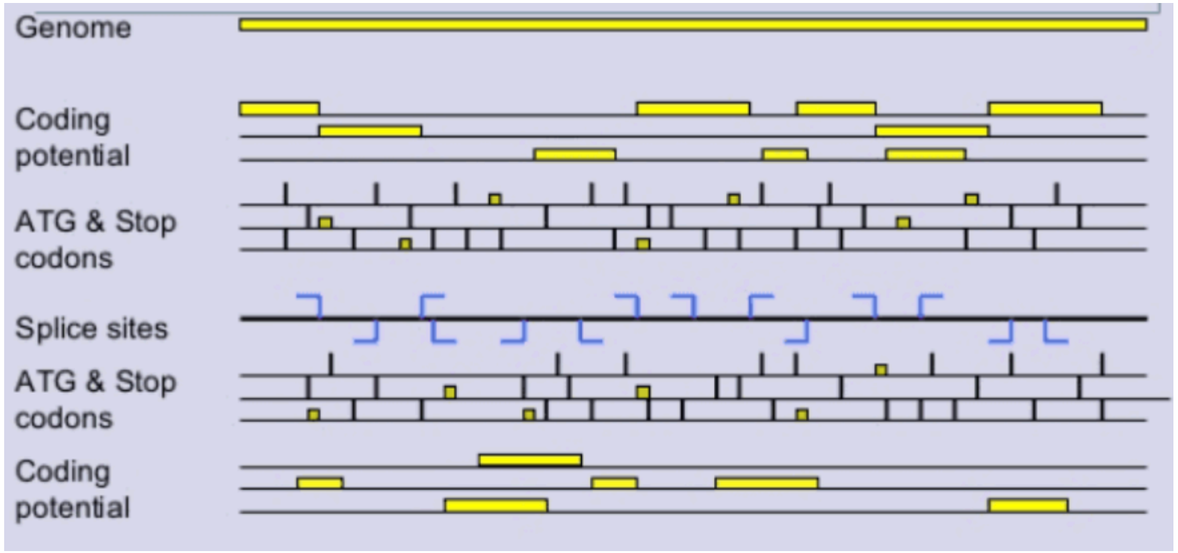
Genome sequence

Statistical models based on the organism of interest

Predictive models mostly based on this guy



The software makes an initial prediction, selects the genes in which it has most confidence, and uses those genes to "retrain" and improve their accuracy in the rest of the genome.
The structure of the genome you´re studying will change the model you need.



*This model was trained with eukaryotic genes, what considerations would a prokaryotic model have?



Augustus for eukaryotes

GLIMMER
Microbial Gene-Finding System

Glimmer for prokaryotes

HOW TO UNIFY PREDICTION AND EVIDENCE?

Data Reconciliation

Imagine that you are annotating a genome and have three different sources of information about the same DNA fragment:

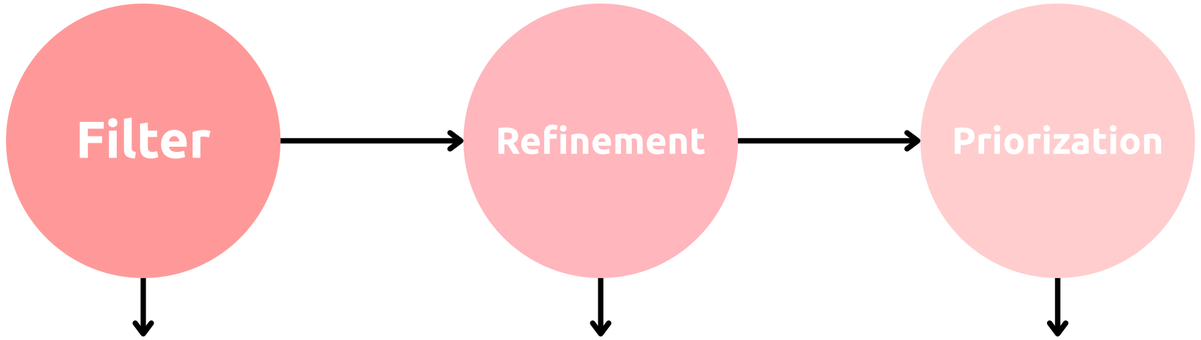
Augustus (Ab initio): Suggests there is a gene with 4 exons based on statistics.

BLAST: Finds a homologous protein whose alignment against the genome implies 5 exons.

RNA-seq: Shows RNA readings covering only part of that region (gene is poorly expressed or an intron is being retained)

Who is the one we believe in?

A pipeline automates a series of decisions that were previously taken manually:



Discards ab initio predictions that lack evidence support (to avoid false positives).

Adjusts the start and stop sites of a gene based on protein or RNA-seq alignments.

If the RNA-seq evidence of the same species is very strong, the pipeline gives it priority over a statistical prediction.

MAKER

Gaius-Augustus/
BRAKER

Prokka

Manual annotation and considerations

Another way of verifying your annotation is recruit experts of some gene families and make a manual curation of their favorite genes



- You get a better annotation
- Definitely WAY pricier
- There aren’t experts for all genes
- They can only annotate what is in the sequence

Apollo

sanger-pathogens/
Artemis

Repeated elements

Before annotating genes, it is mandatory to use ***RepeatMasker***. This "covers" or masks transposons and repetitive sequences. If it is not done, the annotation software gets confused and generates thousands of fake genes.

Exotic elements

Specialized tools are used to find what the standard software ignores, such as tRNAs or rRNAs

Aragorn and tRNAscan

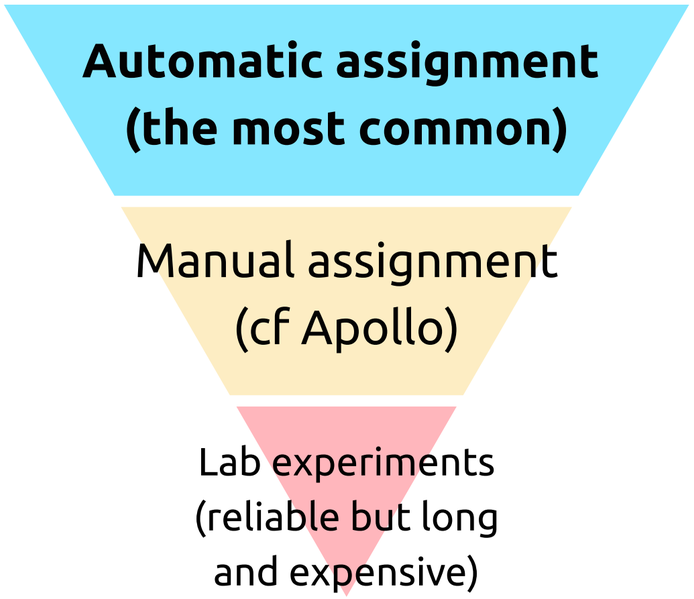
tRNAscan-SE
Searching for tRNA genes in genomic sequence

FUNCTIONAL ANNOTATION

How to assign function

Now that we have the genetic architecture, we need to give these structures a function. A collection of information is made about the function of identified genes: biological function, regulation, expression, etc.

Three allocation sources, from less to more reliable:



The main tools

Blast/Diamond against NR: looks for similar proteins in other species. Risk: Propagates weak "putative protein" annotations of species by species.

InterProScan: Looks for known protein domains and patterns in InterPro. Some manually curated. Delivers Gene Ontology* terms directly.

InterPro

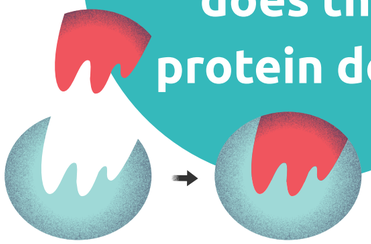
EggNOG-mapper: compares against more than 4 million ortology groups. Assigns name of the gene and function based on which equivalent genes do in other species.

NOG

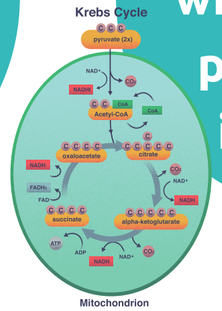
What is gene ontology?

Gene Ontology (GO) is a conceptual framework and international standard of bioinformatics designed to unify the way we describe the function of genes and their products (proteins) in all organisms.

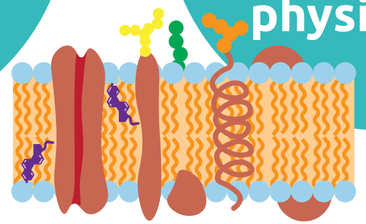
Molecular function (what does the protein do?),



Biological process (in which cellular process is it involved?)



Cellular component (where is it physically?)



- biological_process 808026
- cellular_component 736306
- molecular_function 772551
  - acetylcholine receptor activator activity 14
  - acetylcholine receptor inhibitor activity 38
  - acetylcholine receptor regulator activity 80
  - antioxidant activity 4878
  - binding 309122
  - calcium channel regulator activity 327
  - catalytic activity 368769
  - channel activator activity 25

QUALITY CONTROL

Basic metrics: is my annotation reasonable?

Metric	Measures	A sign of a successful annotation
Total genes	How many genes did the pipeline predict across the genome	Comparable to related species. Prokaryotes: 1,000-10,000 Eukaryotes: 10,000-30,000
Coding density	How "packaged" are the genes in the genome	Consistent with the type of organism.
Average gene length	Average size of each gene (including introns in eukaryotes)	Eukaryotic genes longer by introns
Average length of CDS	Encoding region size (exons only)	Similar to well-annotated proteomes of the same lineage
Average number of exons per gene	Average structural complexity of genes (eukaryotes only)	According to taxonomical group
Genes with an assigned function	Fraction of genes with at least one functional annotation (BLAST, InterPro, GO)	High % of genes with hits in known databases

The gold standard tools

BUSCO: Yes... again we use BUSCO. It looks for single copy orthologous genes that are expected in all organisms of a clade (insects, plants, bacteria...). If your annotation is good, you should find them.

BUSCO

Compleasm: Does exactly the same as BUSCO but with a redesigned algorithm that is significantly faster and, in some clades, more sensitive. The results are comparable and interpreted equally.

OMArk: adds a dimension that BUSCO does not have: Use k-mers composition to assign each protein to hierarchical ortology groups (HOGs) from the OMA database. Then evaluate two things: completeness and phylogenetic consistency



Thanks !!!!!